

DTSA 5742

Predicting Extreme Climate Behavior with Machine Learning

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (62.8 / 100)	52.7 / 100
SPECIALIZATION Climate Change: Time Series Analysis		

53

CURRENT AVG SCORE
across 6 job roles

86

POTENTIAL SCORE
with recommended additions

+36

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course reviews current global climate policies with the goal of gathering data and applying machine learning algorithms to predict extreme climate behaviors, specifically in developing countries. Topics include multivariate linear regression, time-series analysis, and numerical weather prediction. The use of monte carlo simulations to forecast extreme weather events will be analyzed. Strong emphasis will be placed on application in the Python programming language.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	62.8	97.0	+34.2	HIGH
AI Engineer	58.8	94.4	+35.7	HIGH
Data Scientist	55.5	89.1	+33.6	HIGH
ML Engineer	55.3	90.6	+35.3	HIGH
Applied Scientist	49.9	84.1	+34.2	HIGH
MLOps Engineer	33.9	62.1	+28.3	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	62.8 → 97.0 (+34.2 pts)
interfaces predictive tools exploratory design development science models	
AI Engineer	58.8 → 94.4 (+35.7 pts)
models build technical model systems design product new	
Data Scientist	55.5 → 89.1 (+33.6 pts)
product mitigation different build	
ML Engineer	55.3 → 90.6 (+35.3 pts)
systems models model build product new technical building	
Applied Scientist	49.9 → 84.1 (+34.2 pts)
models technical energy systems design development regulatory science	
MLOps Engineer	33.9 → 62.1 (+28.3 pts)
model systems tools models development	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.